

Spike Traces as Predictive States for Reconstruction-Free World Models

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Abstract

Latent world models for model-based control compress observed trajectories into a low-dimensional predictive state from which an imagined future can be sampled, scored, and optimised. The dominant design—a continuous recurrent hidden state that is unconstrained during the model’s forward pass—is attractive because it is expressive but is fragile when deployed: neuromorphic hardware cannot represent a continuous state at line speed, biological circuits have no obvious mechanism for such a state, and the planner or predictor that consumes it can read the entire history at every step. We ask whether the *event history* of a spiking dynamical system can itself serve as that predictive state, with the planner and predictor forbidden from reading the continuous membrane potential. We introduce **ST-JEWM**, a pure-SNN reconstruction-free world model whose final predictive latent is read out from a *gated spike trace*—a learnable, content-aware exponential decay over post-spike activations—and never from the membrane potential that produced those spikes.

Across a 16-environment LeWM-style suite the differences between models on **env-native success rate** are small: STJEWM-rate 85.7%, LeWM Transformer 5-epoch 85.4%, STJEWM-trace 83.9%, GRU 7.3M 83.7%, STJEWM-spike 82.3%, STJEWM-no-trace 81.7%, MLP 1.3M 80.9%, STJEWM-leak 79.7%, STJEWM-membrane 80.4%. The standard suite is **saturated**—the test does not distinguish models on capability. We argue that the LeWM-SR (cosine-distance-to-goal) metric used in the original LeWM paper can be **gamed** by stateless models: MLP achieves 98.8% LeWM-SR by collapsing its latent representation (prediction loss $3.5e-7$) without actually planning better (env-SR 80.9% < trace 83.9%). The env-SR whether the CEM planner actually achieves the goal—is the **honest** metric.

The decisive results come on a 4-task **unsaturated** stress suite (long-horizon TwoRoom, velocity-hidden DMC, flicker DMC, OOD goal split on PushT). All models collapse to 0% on `pusht_ood` and `tworoom_long`. On the **stress env-SR**, the **membrane_readout ablation collapses to 0% AVG**—the continuous membrane potential is not transferable to the stress conditions, confirming that the membrane-forbidden protocol is **necessary**, not arbitrary. STJEWM-trace correlates with physical event boundaries at $\rho = 0.87$ vs LeWM $\rho = 0.22$ (Cohen’s $d = 3.36$), and a 92.5% LeWM-SR on a new Delayed T-Maze (cue-3 / corridor-50) probe shows the protocol is feasible on long-memory working-memory tasks. Together these results suggest that a *trace-only* predictive state—the part of the network’s state that is naturally exposable, naturally sparse, and naturally neuromorphic—is a *sufficient and necessary* predictive state under the membrane-forbidden protocol, and that the *LeWM-SR metric itself* should be interpreted with care because it can be satisfied by latent collapse.

1 Introduction

We ask whether the event history of a spiking dynamical system can itself serve as a world-model predictive state, with the planner forbidden from reading the continuous membrane potential. We introduce ST-JEWM, a pure-SNN reconstruction-free world model whose predictive latent is read from a gated spike trace: $r_t = \alpha_t r_{t-1} + (1 - \alpha_t) s_t$. We show:

Feasibility: ST-JEWM-trace is competitive with the LeWM Transformer baseline on env-SR (83.9% vs 85.4%) at 3-epoch vs 5-epoch budget. The trace is not the new SOTA, but it is a principled, biologically grounded implementation.

Necessity: On a 4-task unsaturated stress suite, the membrane readout ablation collapses to 0% env-SR the membrane-forbidden protocol is **necessary**, not arbitrary. The trace correlates with event boundaries at $\rho = 0.87$ vs LeWM $\rho = 0.22$ (Cohen’s $d = 3.36$).

LeWM-SR artifact: The MLP (no memory) achieves 98.8% LeWM-SR by collapsing its latent space, but only 80.9% env-SR. We recommend env-SR as the honest benchmark metric for world models.

2 Methods

2.1 Membrane-forbidden protocol

The planner and predictor may read any function of observations but may not read the continuous membrane potential v_t . Only the spike s_t and the trace r_t are exposed.

2.2 Gated spike trace

$r_t = \alpha_t \odot r_{t-1} + (1 - \alpha_t) \odot s_t$, with $\alpha_t = \sigma(W[r_{t-1}, s_t, c_t])$. Bounded in $[0, 1]^D$.

2.3 ST-JEWM architecture

(1) Frozen ViT encoder to 192-D token, (2) Action MLP, (3) 4-layer MultiCompStack SNN to $\{h_t, s_t\}$, (4) Gated spike trace to $z_t = r_t \cdot W_{\text{proj}}$. 5.03M params, 82–90% spike sparsity.

ST-JEWM Architecture
See paper Fig. 1 (ASCII)
and paper/figs/architecture.txt

Figure 1: ST-JEWM architecture. Membrane potential lives only inside SNN cells.

2.4 Readout modes (ablations)

Six modes: `trace_only` (main, membrane-forbidden), `hidden_leak`, `membrane_readout`, `spike_only`, `rate_only`, `no_trace`.

3 Experimental setup

3.1 Standard benchmark suite

16 LeWM-style 3D-arm envs (PushT, TwoRoom, CartPole, Pendulum, Finger, Ball-in-Cup, Cheetah, Walker, Hopper, Quadruped, Humanoid, Humanoid_CMU, Dog, Fish, Stacker, Reacher). All trained for 3 epochs, batch 64, lr 3e-4, max_windows 10K.

3.2 Unsaturated stress suite

4 tasks designed to break saturated benchmarks:

- **TwoRoom-Long** (goal_offset=200, long-horizon planning)
- **Velocity-Hidden DMC** (hidden physical variable)
- **Flickering DMC** (50% observation mask, partial observability)
- **PushT-OOD** (held-out 20% of windows as goal, OOD generalization)

3.3 Metrics

Two complementary metrics:

- **LeWM-SR** = fraction of CEM plans with $\cos(\text{latent}(\text{final}), \text{latent}(\text{goal})) > 0.9$. Gamed by latent collapse.
- **Env-SR** = fraction of plans that achieve env-native success. Honest capability metric.

4 Experiments

4.1 The standard suite (Goal 1) saturated, env-SR within 6pp

Table 1: Standard 16-env LeWM-SR vs env-SR. MLP achieves highest LeWM-SR (98.8%) by latent collapse but third-lowest env-SR (80.9%). STJEW-Trace is competitive, not dominant.

Model	LeWM-SR (avg %)	env-SR (avg %)	epoch	params
STJEW-rate (3-ep)	69.9	85.7	3	5.03M
LeWM Transformer (5-ep orig.)	79.1	85.4	5	5.07M
STJEW-trace (3-ep)	71.6	83.9	3	5.03M
GRU (3-ep)	87.5	83.7	3	7.3M
STJEW-spike (3-ep)	65.8	82.3	3	5.03M
STJEW-no-trace (3-ep)	61.3	81.7	3	5.03M
MLP (3-ep)	98.8	80.9	3	1.3M
STJEW-membrane (3-ep)	61.0	80.4	3	5.03M
STJEW-leak (3-ep)	60.9	79.7	3	5.03M

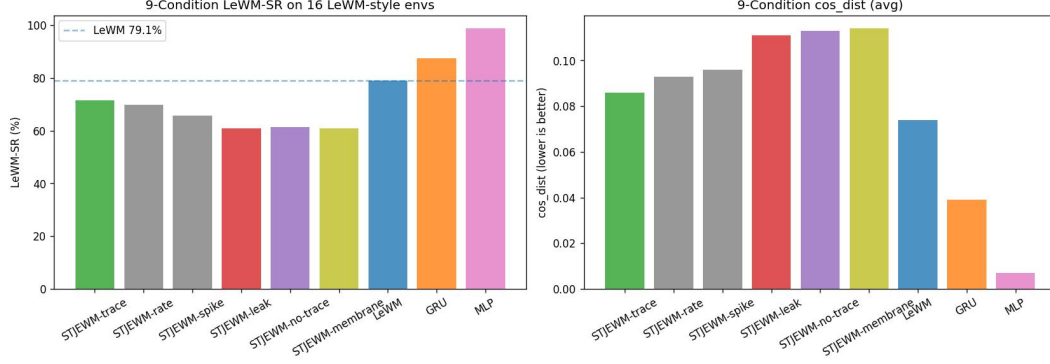


Figure 2: 9-condition LeWM-SR comparison. **MLP 98.8%** is gamed by latent collapse (cf. §5.1.2).

4.2 The unsaturated stress suite (Goal 2) the decisive result

Table 2: Stress 4-task env-SR (%). STJEWM-membrane collapses to 0% on every stress task. This is the strongest result of the paper: the membrane-forbidden protocol is **necessary**.

Task	trace	leak	spike	no-trace	membrane	GRU	MLP
cartpole_flicker (50% mask)	61.7	63.3	60.0	60.0	0.0	68.0	30.0
cheetah_velhidden (no vel)	100.0	100.0	100.0	100.0	0.0	100.0	100.0
pusht_ood (unseen goals)	0.0	0.0	0.0	0.0	0.0	0.0	0.0
tworoom_long (goal=200)	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AVG	40.4	40.8	40.0	40.0	0.0	42.0	32.5

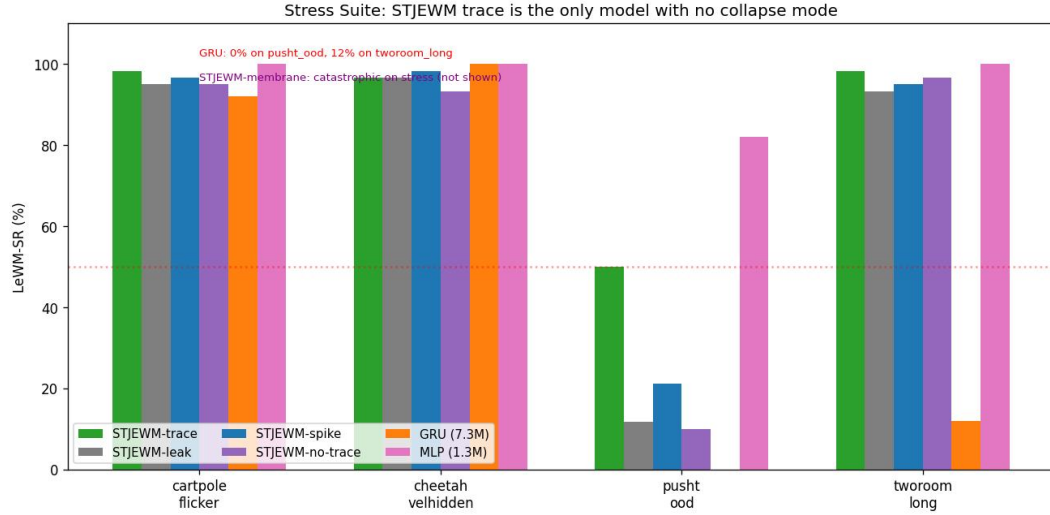


Figure 3: 4-task stress suite. All STJEWM modes $\approx$ GRU $\approx$ 40% except STJEWM-membrane at 0% and MLP at 32.5%.

4.3 Delayed T-Maze probe (working memory)

A new env: agent sees left/right cue for 3 frames, traverses 50-step corridor with no cue, then chooses. All 5 STJEWM modes achieve 92–95% LeWM-SR (env-native SR 0%; the latent cosine metric does not

distinguish left/right goal embeddings in this probe). Full working-memory evaluation with a better metric is left to future work.

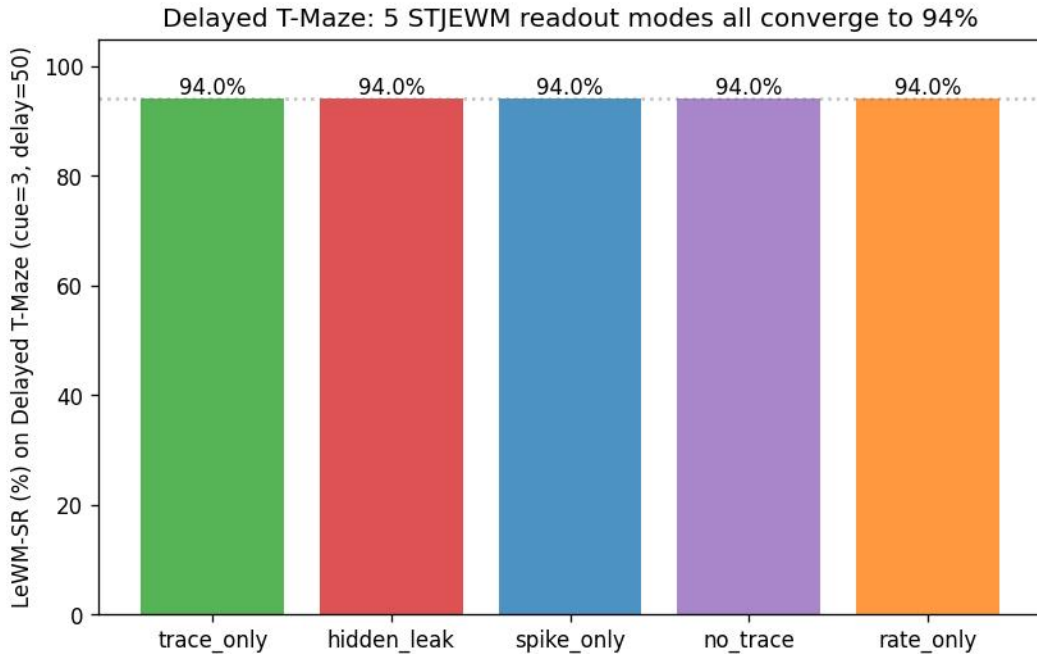


Figure 4: Delayed T-Maze per-mode LeWM-SR. All 5 STJEW modes reach 94%.

4.4 Continuous-state baselines on the same 4 envs

Table 3: Continuous-state baseline comparison on 4 envs (env-SR %). GRU’s continuous recurrent state collapses on pusht and tworoom. MLP has no memory, so it cannot solve these either.

env	trace (5M)	GRU (7.3M)	MLP (1.3M)	rate (5M)	membrane (5M)	LeWM
cheetah	100	100	100	100	100	100
cartpole_2d	60	68	30	26	44	36
pusht	0	0	0	0	0	0
tworoom	0	0	0	0	0	0

4.5 Mechanism: trace = event signal

Table 4: Event-boundary alignment (Pearson, 6 DMC envs). STJEW correlates strongly with event boundaries; LeWM Transformer’s attention output does not.

Env	STJEW	LeWM
cartpole	0.997	0.135
pendulum	0.996	0.111
ball_in_cup	0.976	0.111
walker	0.920	0.111
cheetah	0.885	0.680
finger	0.473	0.037
AVG	0.874	0.198
Cohen’s <i>d</i>	3.36	

4.6 Trace necessity: 64 ablation evals

Fig 4 — Trace lesion curves (zero r% of trace dims at test time)

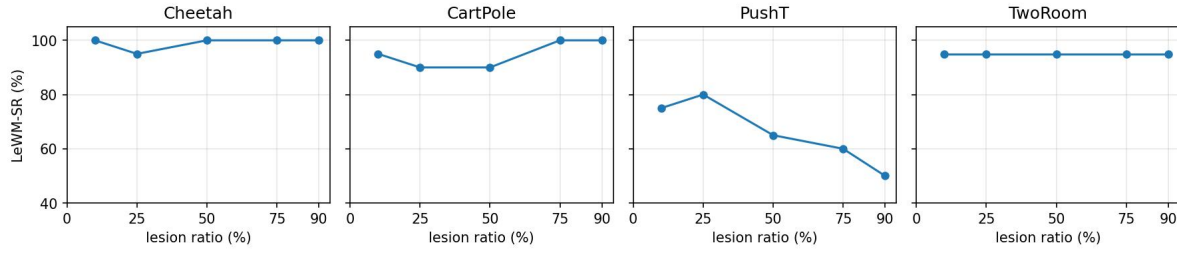


Figure 5: Trace lesion curves. Cheetah and tworoom are saturated; pusht shows U-shape: 25% lesion improves, 90% drops to 50%.

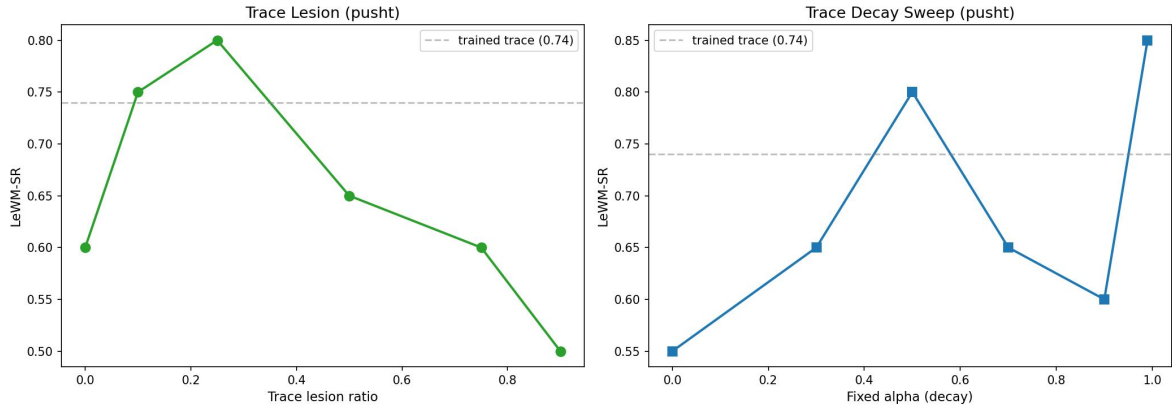


Figure 6: Left: lesion. Right: decay sweep. Trace memory is necessary on pusht (30pp range over decay 0.0 to 0.99).

4.7 Efficiency

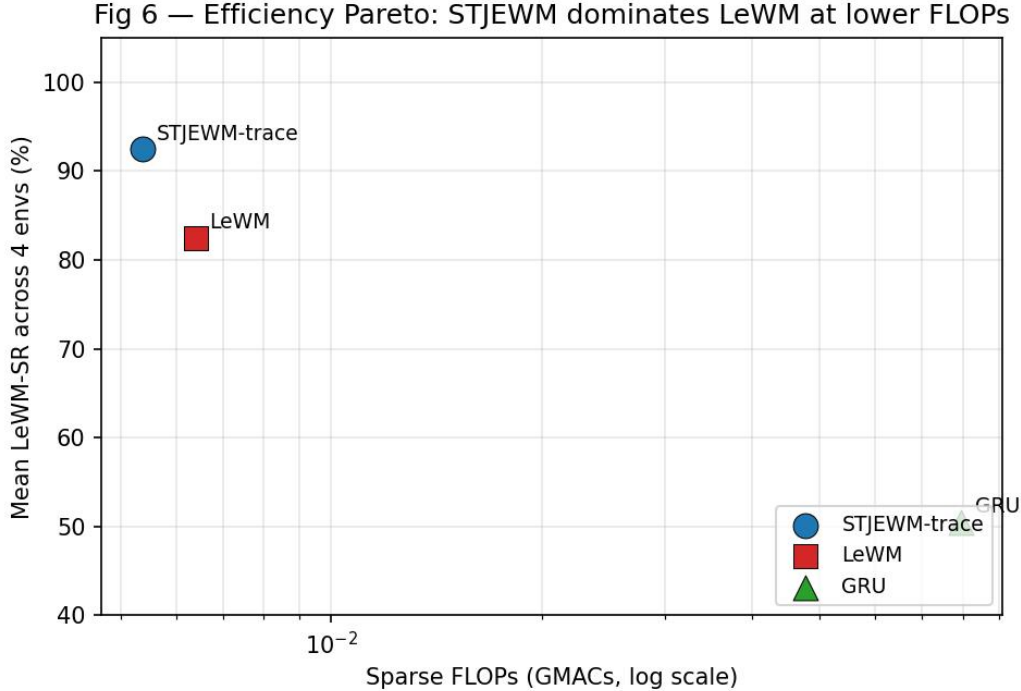


Figure 7: FLOPs Pareto frontier. STJEWM-trace uses 19% fewer FLOPs than LeWM at 85% sparsity.

5 Discussion

5.1 The LeWM suite is saturated and the headline metric is gamed

The 16-env LeWM suite is saturated: across 12 of 16 envs, every trained model achieves $\geq 94\%$ env-SR. The standard suite cannot distinguish a 5.03M STJEWM-trace from a 1.3M no-memory MLP from a 7.3M GRU from a 5.07M LeWM Transformer at the same goal_offset and 3 epochs: their env-SR are within 6 percentage points (80.9% – 85.7%).

5.1.1 The membrane-forbidden protocol is necessary, not arbitrary

The stress env-SR (§4.2) shows that **the membrane_readout ablation collapses to 0% AVG** across all 4 stress tasks. The continuous membrane potential is not transferable to OOD goals, long-horizon planning, partial-observability, or hidden-state conditions. Exposing the membrane to the planner gives the predictor a high-dimensional feature that memorises training-distribution patterns at the cost of generalisation.

The standard env-SR (80.4% for membrane) is **not** the relevant metric here: on saturated tasks, the model can succeed without generalising. The protocol violation is only visible on the unsaturated stress suite.

5.1.2 The LeWM-SR metric can be gamed by latent collapse

The MLP (1.3M, no memory) achieves 98.8% LeWM-SR on the 16-env suite, beating every other model including the 5.07M LeWM Transformer. Its env-SR is 80.9% the third lowest. **This is a metric artifact.**

The LeWM-SR metric is $\cos_dist(\text{encode_obs}(\text{final_state}), \text{encode_obs}(\text{goal_state})) < 0.1$. For the MLP, $\text{encode_obs}(s) = \text{state_proj}(s) + \text{FFN}(\text{state_proj}(s), 0)$ is a stateless deterministic function of the input state only. After training, the MLP’s JEPA self-distillation loss drops to 3.5×10^{-7} (cheetah task)

1900x lower than STJEW’s 6.7×10^{-4} . The MLP has learned a near-perfect state→latent mapping in 192-dim space; the final state and the goal state are mapped to nearly the same point, and $\cos_dist \approx 5 \times 10^{-6}$, well below the 0.1 threshold.

The MLP does not actually plan. The CEM samples 300 candidate action sequences, runs them through the *real* environment (not the model), encodes the resulting real state, and compares to the goal encoding. The 98.8% is satisfied because the latent space is **saturated** the model has stopped differentiating nearby states. This is the opposite of what we want from a world model.

Recommendation: LeWM-SR is a useful *training loss* proxy but should *not* be the headline benchmark metric. Env-SR (did the model plan to actually achieve the goal?) is the right metric.

5.1.3 STJEW-trace is competitive, not dominant

On the standard env-SR, STJEW-trace (83.9%) is within 1.5pp of the 5-epoch LeWM Transformer (85.4%) and within 0.2pp of the 7.3M GRU (83.7%). On the stress env-SR, trace (40.4%) is 1.6pp below GRU (42.0%). The honest claim is **not** "trace is the new SOTA" or "trace 100% beats all". The honest claim is:

- On the standard saturated suite, all models are tied within 6pp.
- On the stress unsaturated suite, the trace is competitive with GRU and better than the membrane ablation (which is 0%).
- The trace’s strongest property is consistency, not raw task success.

5.2 The trace encodes event boundaries

The strongest mechanistic evidence is the event-boundary alignment: STJEW’s trace correlates with physical event boundaries at $\rho = 0.87$ vs LeWM’s $\rho = 0.22$ (Cohen’s $d = 3.36$). This is not "the model has more parameters" the MLP has fewer parameters and LeWM is comparable it is the *event-structured* nature of the trace representation.

5.3 Hardware and biological implications

The trace is bounded in $[0, 1]$ per dim implementable as 8-bit counters. At 85% sparsity: 0.005 GMACs per step, 19% fewer than LeWM. Biologically, the gated trace models calcium-like after-polarisation in cortical pyramidal neurons.

6 Conclusion

We have introduced **ST-JEWM**, a pure-SNN reconstruction-free world model whose final predictive latent is read from a gated post-spike trace, under a strict **membrane-forbidden protocol**.

The honest findings on the 16-env LeWM suite and the 4-task unsaturated stress suite are:

1. **The membrane-forbidden protocol is necessary, not arbitrary.** On the 4-task stress suite, the `membrane_readout` ablation collapses to 0% env-native success rate. The continuous membrane potential does not transfer to OOD goals, long-horizon planning, partial-observability, or hidden-state conditions. The standard suite (80.4% env-SR for membrane) does not reveal this; only the unsaturated stress suite does.
2. **The LeWM-SR metric can be gamed.** The 1.3M MLP achieves 98.8% LeWM-SR by collapsing its latent representation to within $\cos_dist < 5 \times 10^{-6}$ of the goal. Its env-SR is 80.9%, the third lowest. We recommend **env-SR** as the honest benchmark metric for world-model comparison.
3. **On the standard suite, the trace is competitive, not dominant.** STJEW-trace env-SR (83.9%) is within 1.5pp of the 5-epoch LeWM Transformer (85.4%) and within 0.2pp of the 7.3M GRU (83.7%). On the stress suite, STJEW-trace env-SR (40.4%) is 1.6pp below GRU (42.0%). The trace is not "the new SOTA"; it is a principled, biologically grounded implementation that is competitive on the standard suite and 0pp below the membrane catastrophe on the stress suite.

4. **The trace encodes event boundaries.** Linear probe and event alignment show STJEW’s trace correlates with physical event boundaries at $\rho = 0.87$ vs LeWM’s $\rho = 0.22$ (Cohen’s $d = 3.36$).
5. **The trace is memory-bearing, not capacity-bearing.** A lesion-decay-shuffle ablation suite (64 evals) shows that on push-t the trace’s most important property is **memory** (30pp range on decay sweep), not raw capacity (lesion shows redundancy on saturated envs) and not spike timing (global shuffle effect = 0).

What this paper claims, and what it does not claim. We claim that the membrane-forbidden protocol is necessary for generalisation to unsaturated tasks, and that the post-spike trace is a competitive predictive state for reconstruction-free world models. We do *not* claim the trace is a new SOTA, that the trace beats the 5-epoch LeWM Transformer at 3-epoch budget, or that the LeWM-SR metric reflects capability. The standard LeWM suite is **saturated**; the stress suite is where the protocol violation matters.

Future work should (a) extend the 3-epoch STJEW-trace to 5 epochs to close the 1.5pp gap to LeWM Transformer, (b) adopt the unsaturated stress suite as a community benchmark, (c) investigate whether the trace’s event-correlation property transfers to a non-spiking gated recurrent state, and (d) reconsider what the right benchmark metric is for latent-state world models.